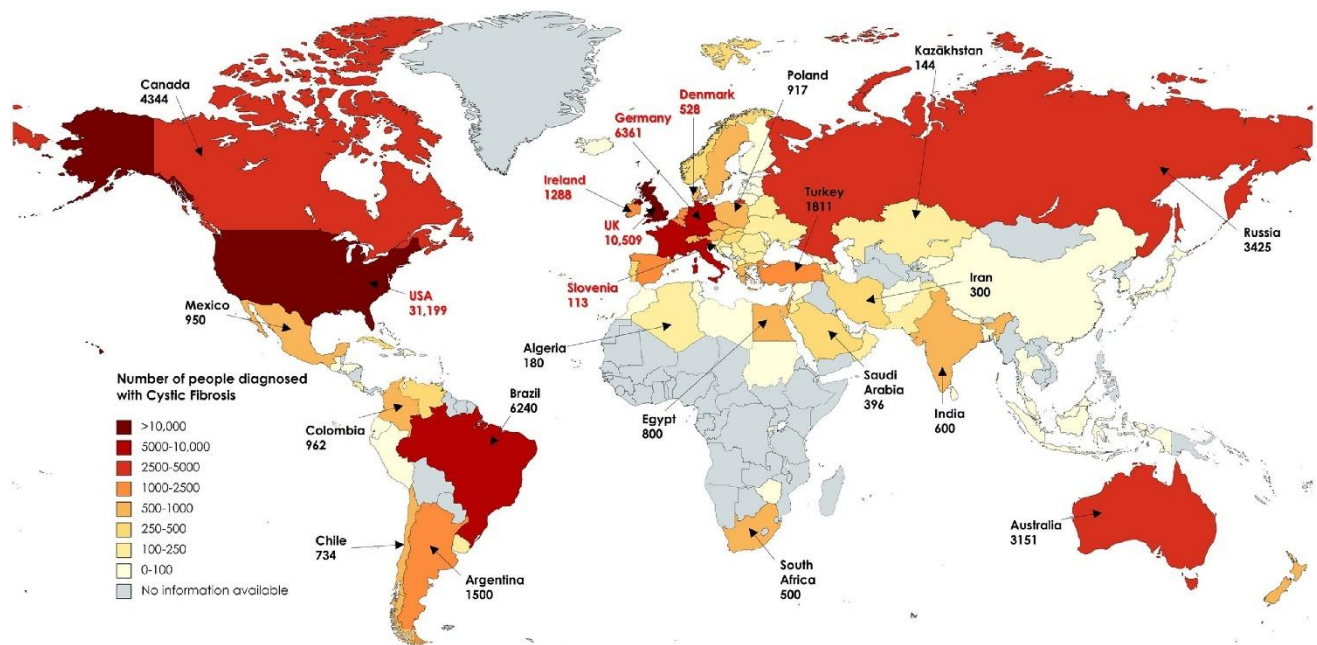


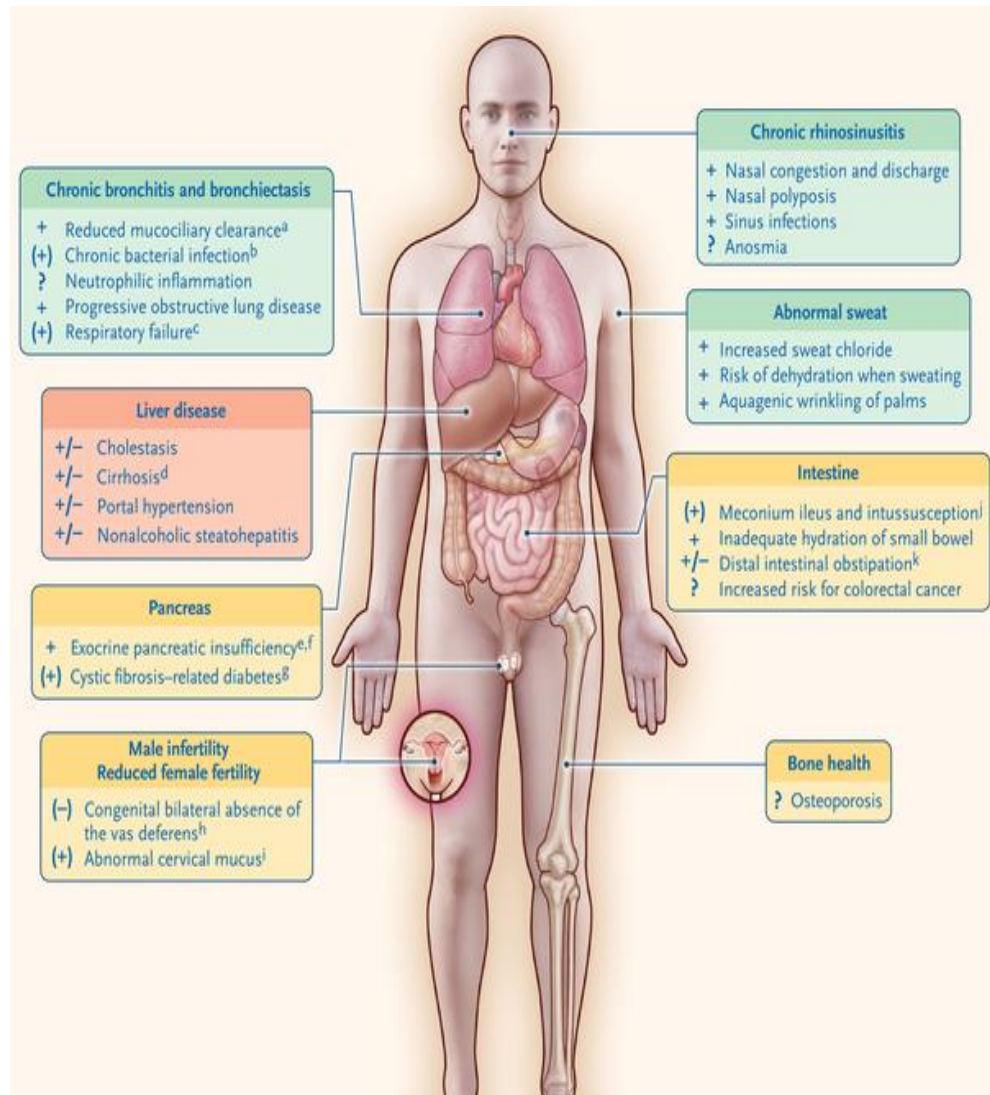
- SCN5A, SCN10A, SLMAP and TRPM4*
 - Potassium current (I_K): *ABCC9, KCND2, KCND3, KCNE1L, KCNE3, KCNE5, KCNH2, KCNJ8, SEMA3A*
 - Calcium current (I_{Ca}): *CACNA1C, CACNA2D1, CACNB2B*
 - Miscellaneous/Other: *HCN4, HEY2, LRRC10*
- Some Drugs to be avoided** (brugadadrugs.org):
 - Antiarrhythmic drugs*: Ajmaline, Allapinine, Ethacizine, Flecainide, Pilsicainide, Procainamide, Propafenone
 - Psychotropic drugs: Amitriptyline, Clomipramine, Desipramine, Lithium, Loxapine, Nortriptyline, Oxcarbazepine, Trifluoperazine
 - Anesthetics / analgesics*: Bupivacaine, Procaine, Propofol
 - Other substances: Acetylcholine, Alcohol (toxicity), Cannabis, Cocaine, Ergonovine

Slide – 29: Cystic Fibrosis

- **Map displaying the estimated diagnosed CF patient burden around the world** (Journal of Cystic Fibrosis Volume 21, Issue 3, May 2022, Pages 456-462)



Slide – 30: Cystic Fibrosis – Clinical Features (N Engl J Med 2023;389:1693-1707):



Slide – 32: Cystic Fibrosis Carrier Screening

Basic and expanded screening panels for cystic fibrosis

**Standard or basic pathogenic variant panel**

F508del-CFTR, I507del, G542X, G551D, W1282X, N1303K, R553X, 621+1G>T, R117H, 1717-1G>A, A455E, R560T, R1162X, G85E, R334W, R347P, 711+1G>T, 1898+1G>A, 2184delA, 3849+10kbC>T, 2789+5G>A, 3659delC, 3120+1G>A

Reflex tests

I506V*, I507V*, F508C*, 5T/7T/9T¶

CF: cystic fibrosis.

* Benign variants. This test distinguishes between the F508del-CFTR pathogenic variant and these benign variants. I506V, I507V, and F508C testing is performed only as reflex tests for unexpected homozygosity for F508del-CFTR and/or I507del.

¶ 5T/7T/9T analysis is performed only when R117H is detected for the following reasons: The R117H variant can result in classic CF only if there is a 5T variant on the same chromosome and another CF variant on the opposite chromosome. R117H on 1 chromosome and 5T or 7T on the opposite chromosome or homozygosity for 5T (5T on each chromosome) can result in congenital bilateral absence of the vas deferens.

Data from:

1. American College of Medical Genetics. Technical standards and guidelines for CFTR mutation testing. 2006.
2. Committee on Genetics, American College of Obstetricians and Gynecologists. ACOG Committee Opinion. Number 325, December 2005. Update on carrier screening for cystic fibrosis. Obstet Gynecol 2005; 106:1465.
3. Castellani C, Cuppens H, Macek M Jr, et al. Consensus on the use and interpretation of cystic fibrosis mutation analysis in clinical practice. J Cyst Fibros 2008; 7:179.

Graphic 65637 Version 9.0

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Slide – 33: CF Treatment**Medications**

- **Mucolytics**: Dornase alfa/DNAse, Hypertonic Saline
- **Bronchodilators**: Albuterol
- **Antibiotics**: Oral, IV, Inhalation
- **CFTR Potentiators**: Ivacaftor (Kalydeco) for CFTR mutations where CFTR protein is present but Cl- gate “locked”→ Opens locked channel (38 FDA-approved mutation genotypes)
- **CFTR Modulators**: Lumacaftor (Orkambi) for CFTR mutations where protein stuck in Endoplasmic Reticulum (ER)→ Increases transport of protein out of the ER (F508del Homozygotes)